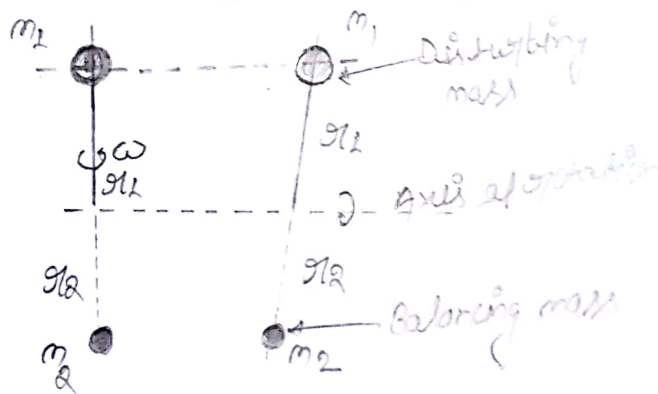


Unit - IV

Balancing: Static and dynamic balancing - balancing of rotating masses, single and multi-cylinder engines. Reciprocating masses - single cylinder engine.

- * **Dynamic balancing:** When several masses rotate in different planes, the centrifugal forces, being out of balance, also form couples.
 "A system of rotating masses is in dynamic balance when there does not exist any F_c as well as couple."
- * **Static Balance:** A system of masses is said to be in static balance if the combined mass centre of the system lies on the axis of rotation.

"Balancing of a single rotating mass by a single mass in the same plane"



$$F_{C1} = m_1 r_1 \omega^2 \quad \text{--- (i)}$$

$$F_{C2} = m_2 r_2 \omega^2 \quad \text{--- (ii)}$$

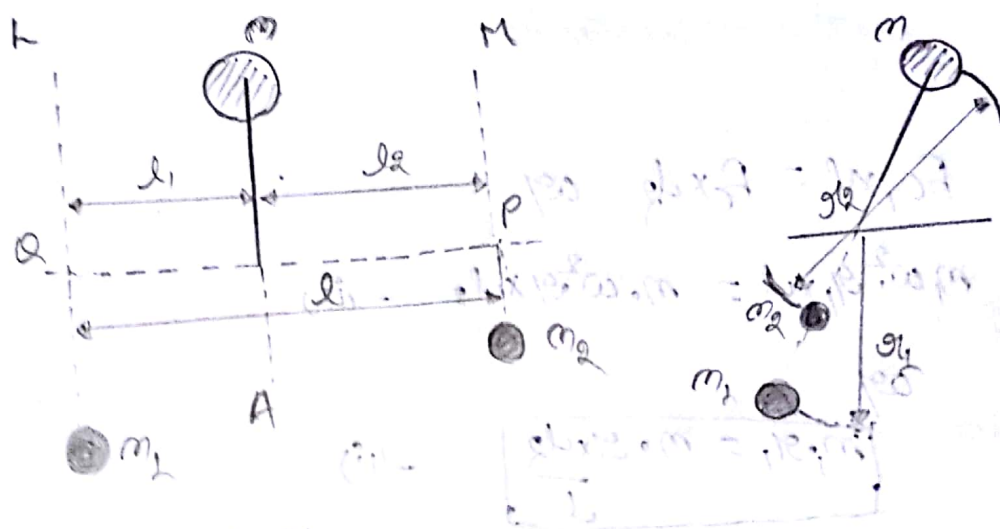
$$F_{C1} = F_{C2}$$

$$m_1 r_1 \omega^2 = m_2 r_2 \omega^2$$

$$\boxed{m_1 r_1 = m_2 r_2}$$

{ r_1 = Radius of rotation of the disturbing mass m_1
 r_2 = distance b/w the axis of rotation of the shaft and centre of gravity of mass m_2 }

Balancing of a single rotating mass by two rotating masses in different planes



A, M, L are planes

l_1 = distance b/w planes A and L
 l_2 = " " " " M and L
 l_3 = " " " " M and L

$$F_c = m\omega^2 r$$

$$F_{c1} = m_1 \omega^2 r_1$$

$$F_{c2} = m_2 \omega^2 r_2$$

For balancing, the sum of the centrifugal forces must be zero.

$$F_c = F_{c1} + F_{c2} = 0$$

$$m r = m_1 r_1 + m_2 r_2$$

In plane 2)
 (dynamic force at the bearing P of a shaft)

take moment about P which is the point of intersection of the plane M and the axis of rotation. Therefore

$$F_{C1} \times d = F_C \times d_2 \quad \text{or}$$

$$m_1 \omega^2 r_1 \times d = m \omega^2 r \times d_2 \quad \text{--- (i)}$$

$$\text{or} \quad \boxed{m_1 r_1 = m \cdot r \times \frac{d_2}{d}} \quad \text{--- (i')}$$

In plane M
 Similarly, (dynamic force at the bearing Q of a shaft)
 at the bearing P of a shaft

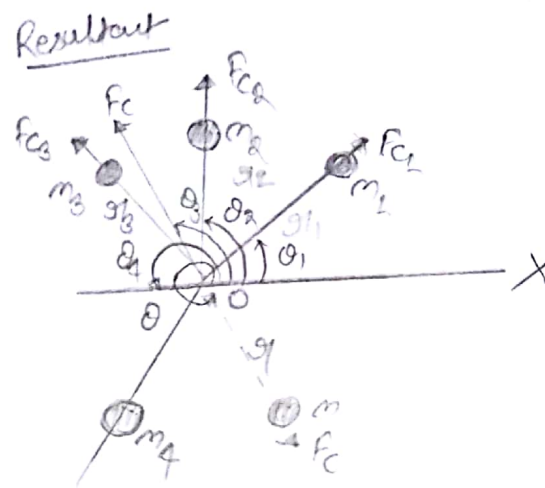
taking moment about Q

$$F_{C2} \times d = F_C \times d_1$$

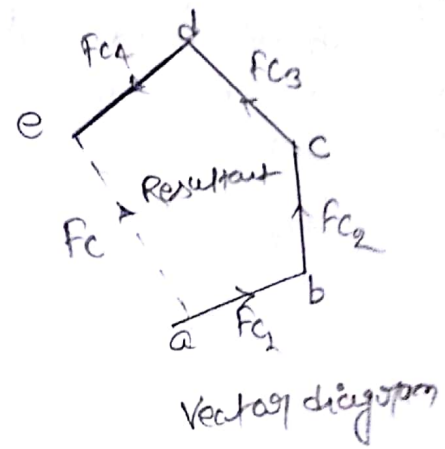
$$m_2 \omega^2 r_2 \cdot d = m \omega^2 r \times d_1$$

$$\boxed{m_2 r_2 = m \cdot r \times \frac{d_1}{d}} \quad \text{--- (ii')}$$

equation (i) for static balance must be satisfied
 and (ii') & (iii') must be satisfied for dynamic balance



(a) Space diagram



Vector diagram

(Balancing of several masses rotating in the same plane.)

2. Analytical Method

- (i) Find out the centrifugal force
- (ii) Resolve the centrifugal force horizontally and vertically and find their sum
i.e. ΣH and ΣV

we know that

$$\Sigma H = m_1 r_1 \cos \theta_1 + m_2 r_2 \cos \theta_2 + \dots$$

$$\Sigma V = m_1 r_1 \sin \theta_1 + m_2 r_2 \sin \theta_2 + \dots$$

$$F_c = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

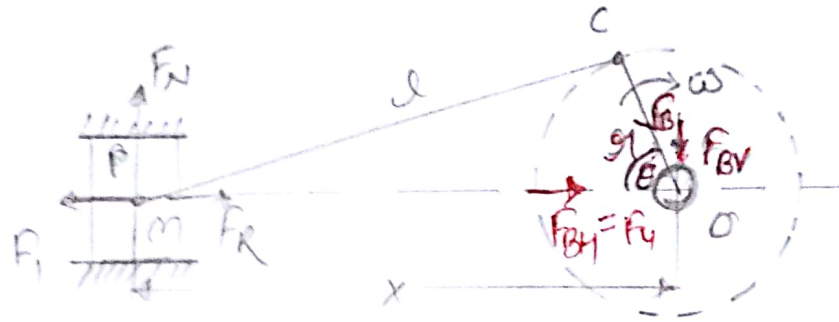
$$\boxed{\tan \theta = \frac{5V}{54}}$$

Note

the balancing force is equal to resultant force, but in opposite direction.

Balancing of Reciprocating masses:

The resultant of all the forces acting on the body of the engine due to inertia forces only is known as "unbalanced force or shaking force".



F_R = force required to accelerate the reciprocating parts

F_L = Inertia force due to reciprocating parts

F_N = forces on the sides of the cylinder walls or normal force acting on the cross-head guides

F_B = force acting on the crankshaft bearing or main bearing

{ Note: The masses rotating with the crankshaft are normally balanced and they do not transmit any unbalanced or shaking force on the body of the engine. }

$m = \text{mass of the reciprocating parts}$
 $l = \text{length of the connecting rod}$

$r = \text{radius of the crank}$

$\theta = \text{Angle of inclination of the crank with the line of stroke } PO,$

$\omega = \text{Angular speed of the crank}$

$$n = \frac{l}{r}$$

Acceleration of the reciprocating parts is

$$a_R = \omega^2 r \left(\cos \theta + \frac{\cos 2\theta}{n} \right)$$

$\therefore$ Inertia force

$$F_I = F_R = m \times a_R = m \cdot \omega^2 \cdot r \left(\cos \theta + \frac{\cos 2\theta}{n} \right)$$

Unbalanced force

$$F_U = m \cdot \omega^2 \cdot r \left(\cos \theta + \frac{\cos 2\theta}{n} \right)$$

$$= m \cdot \omega^2 \cdot r \cos \theta + m \cdot \omega^2 \cdot r \cdot \frac{\cos 2\theta}{n}$$

$$= F_P + F_S$$

primary unbalanced force
 Secondary unbalanced force

Notes:

$F_{P \max} =$ when $\theta = 0^\circ$ or 180° thus primary force is max^m twice in one revolution

$F_{S \max} =$ when $\theta = 0^\circ, 90^\circ, 180^\circ$ and 360° thus secondary force is maximum four times in one revolution

$$F_{P \max} = m \cdot \omega^2 \cdot r$$

$$F_{S \max} = m \cdot \omega^2 \cdot \frac{r}{n}$$

$\therefore$ Secondary force is $1/n$ times the primary force when both are maximum

Partial Balancing of Unbalanced primary force in a reciprocating engine

$$F_{u(primary)} = m \omega^2 r \cos \theta$$

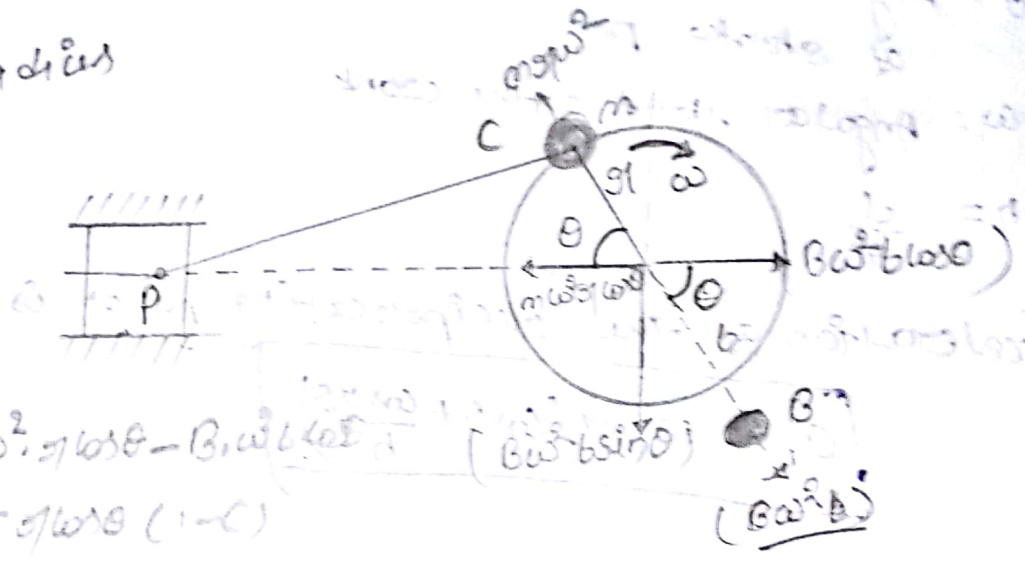
r = crank radius
 m = mass

$$C.m.r = B.b$$

unbalanced force along the line of stroke

$$= m \omega^2 r \cos \theta - B \omega^2 b \sin \theta$$

$$= m \omega^2 r \cos \theta (1 - C)$$



Resultant unbalanced force at any instant

$$= \sqrt{[(1-C)m \omega^2 r \cos \theta]^2 + [C.m \omega^2 r \sin \theta]^2}$$